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ELECTRONIC DEVICE COMPRISING CAMERA MODULE

Abstract

An electronic device comprises: a printed circuit board comprising a ground portion; a camera comprising a lens facing a first direction; a camera housing including an opening in a first surface facing the first direction, and surrounding at least a portion of the camera; and a bracket electrically connected to the ground portion, and at least partially surrounding the camera housing. The camera housing comprises a bridge extending from the camera housing to a third surface of the bracket.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/KR2023/015782 designating the United States, filed on Oct. 13, 2023, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application Nos. 10-2022-0158786, filed on Nov. 23, 2022, 10-2022-0179857, filed on Dec. 20, 2022, and 10-2023-0034217, filed on Mar. 15, 2023, in the Korean Intellectual Property Office, the disclosures of each of which are incorporated by reference herein in their entireties.

BACKGROUND

Field

[0002] The present disclosure relates to an electronic device including a camera module.

Description of Related Art

[0003] An electronic device may include a camera module for photographing a still image and/or a moving image. For example, the electronic device may include a front camera exposed to the front of the electronic device through an opening or a rear camera exposed through a rear cover. The electronic device may include a camera housing surrounding the camera. The camera housing may be grounded to shield electromagnetic waves emitted from the camera and electromagnetic waves reaching the camera housing from other electronic components.

[0004] The above-described information may be provided as a related art for the purpose of helping to understand the present disclosure. No assertion or determination is made as to whether any of the above-described information may be applied as a prior art related to the present disclosure.

SUMMARY

[0005] An electronic device according to an example embodiment may comprise: a printed circuit board, a camera, a camera housing, and a bracket. The printed circuit board may include a ground portion. The camera may include a lens. The lens may face a first direction. The camera housing may include an opening. The opening may be configured for the lens to be exposed toward the outside within a first surface facing the first direction. The camera housing may surround at least a portion of the camera. The bracket may include a third surface. The third surface may face away from a second surface of the camera housing, perpendicular to the first surface. The bracket may be electrically connected to the ground portion. The bracket may at least partially surround the camera housing. The camera housing may include a bridge. The bridge may extend from the camera housing to the third surface of the bracket.

[0006] An electronic device according to an example embodiment may comprise: a printed circuit board, a camera, a camera housing, an injection portion, and a connector. The printed circuit board may include a ground portion. The camera may include a lens. The camera may be disposed on the printed circuit board. The camera housing may surround at least a portion of the camera. The injection portion may be disposed inside the camera housing. The injection portion may surround at least a portion of the camera. The connector may extend from a portion of an inner surface of the camera housing, which is exposed through a groove of the injection portion, to the ground portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The above and other aspects, features and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

[0008] FIG. **1** is a block diagram illustrating an example electronic device in a network environment according to various embodiments;

[0009] FIG. **2** is a block diagram illustrating an example configuration of a camera module according to various embodiments;

[0010] FIG. **3** is an exploded perspective view of an example electronic device according to various embodiments;

[0011] FIG. **4** is a partial exploded perspective view of the electronic device of FIG. **3** according to various embodiments;

[0012] FIG. **5**A is a perspective view illustrating an example camera housing according to various embodiments;

[0013] FIG. **5**B is a diagram illustrating an example process of forming an example camera housing according to various embodiments;

[0014] FIG. **6** is a cross-sectional view of an example electronic device taken along line A-A′ of FIG. **3** according to various embodiments;

[0015] FIGS. **7**A and **7**B diagrams illustrating the camera housing illustrated in FIG. **6** being coupled to a bracket according to various embodiments;

[0016] FIG. **8**A is a cross-sectional view of an example electronic device taken along line A-A′ of FIG. **3** according to various embodiments;

[0017] FIGS. **8**B and **8**C are diagrams illustrating the camera housing illustrated in FIG. **8**A being coupled to a bracket according to various embodiments;

[0018] FIG. **9** is a perspective view illustrating an example camera housing according to various embodiments;

[0019] FIGS. **10**A and **10**B are cross-sectional views of an example electronic device including the camera housing of FIG. **9** taken along line A-A′ of FIG. **3** according to various embodiments;

[0020] FIG. **11**A is a perspective view illustrating an example camera housing being connected to a printed circuit board according to various embodiments;

[0021] FIG. **11**B is a perspective view illustrating an example camera housing being connected to a support according to various embodiments;

[0022] FIG. **11**C is a diagram illustrating an example process of forming an example camera housing according to various embodiments;

[0023] FIG. **12**A is a perspective view illustrating the camera housing illustrated in FIG. **11**A or **11**B being coupled to a bracket in a first direction according to various embodiments;

[0024] FIG. **12**B is a perspective view illustrating the camera housing illustrated in FIG. **11**A or **11**B being coupled to a bracket in a direction opposite to the first direction according to various embodiments;

[0025] FIG. **12**C is a perspective view illustrating the camera housing illustrated in FIG. **10**B being coupled to a bracket in a direction opposite to the first direction according to various embodiments;

[0026] FIG. **13**A is an exploded perspective view of an example camera and camera housing according to various embodiments;

[0027] FIG. **13**B is a perspective view illustrating an injection portion of an example camera housing according to various embodiments;

[0028] FIG. **14**A is a cross-sectional view of an example electronic device taken along line A-A′ of FIG. **3** according to various embodiments; and

[0029] FIG. **14**B is a diagram illustrating an enlarged view of a portion X of FIG. **14**A according to various embodiments.

DETAILED DESCRIPTION

[0030] FIG. **1** is a block diagram illustrating an example electronic device **101** in a network environment **100** according to various embodiments.

[0031] Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In various embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In various embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).
[0032] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function.
[0033] The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**. Thus, the processor **120** may include various processing circuitry and/or multiple processors. For example, as used herein, including the claims, the term “processor” may include various processing circuitry, including at least one processor, wherein one or more of at least one processor, individually and/or collectively in a distributed manner, may be configured to perform various functions described herein. As used herein, when “a processor”, “at least one processor”, and “one or more processors” are described as being configured to perform numerous functions, these terms cover situations, for example and without limitation, in which one processor performs some of recited functions and another processor(s) performs other of recited functions, and also situations in which a single processor may perform all recited functions. Additionally, the at least one processor may include a combination of processors performing various of the recited/disclosed functions, e.g., in a distributed manner. At least one processor may execute program instructions to achieve or perform various functions.
[0034] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**.

According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0035] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0036] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0037] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0038] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0039] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0040] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0041] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0042] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0043] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0044] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0045] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0046] The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0047] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0048] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0049] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device

101, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0050] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0051] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0052] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0053] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0054] FIG. **2** is a block diagram **200** illustrating an example configuration the camera module **180** according to various embodiments.

[0055] Referring to FIG. **2**, the camera module **180** may include a lens assembly (e.g., including at least one lens) **210**, a flash **220**, an image sensor **230**, an image stabilizer (e.g., including circuitry) **240**, memory **250** (e.g., buffer memory), or an image signal processor (e.g., including circuitry) **260**. The lens assembly **210** may collect light emitted or reflected from an object whose image is to be taken. The lens assembly **210** may include one or more lenses. According to an embodiment, the camera module **180** may include a plurality of lens assemblies **210**. In such a case, the camera module **180** may form, for example, a dual camera, a 360-degree camera, or a spherical camera. Some of the plurality of lens assemblies **210** may have the same lens attribute (e.g., view angle, focal length, auto-focusing, f number, or optical zoom), or at least one lens assembly may have one or more lens attributes different from those of another lens assembly. The lens assembly **210** may include, for example, a wide-angle lens or a telephoto lens.

[0056] The flash **220** may emit light that is used to reinforce light reflected from an object. According to an embodiment, the flash **220** may include one or more light emitting diodes (LEDs) (e.g., a red-green-blue (RGB) LED, a white LED, an infrared (IR) LED, or an ultraviolet (UV) LED) or a xenon lamp. The image sensor **230** may obtain an image corresponding to an object by converting light emitted or reflected from the object and transmitted via the lens assembly **210** into an electrical signal. According to an embodiment, the image sensor **230** may include one selected from image sensors having different attributes, such as a RGB sensor, a black-and-white (BW) sensor, an IR sensor, or a UV sensor, a plurality of image sensors having the same attribute, or a plurality of image sensors having different attributes. Each image sensor included in the image sensor **230** may be implemented using, for example, a charged coupled device (CCD) sensor or a complementary metal oxide semiconductor (CMOS) sensor.

[0057] The image stabilizer **240** may include various circuitry and move the image sensor **230** or at least one lens included in the lens assembly **210** in a particular direction, or control an operational attribute (e.g., adjust the read-out timing) of the image sensor **230** in response to the movement of the camera module **180** or the electronic device **101** including the camera module **180**. This allows compensating for at least part of a negative effect (e.g., image blurring) by the movement on an image being captured. According to an embodiment, the image stabilizer **240** may sense such a movement by the camera module **180** or the electronic device **101** using a gyro sensor (not shown) or an acceleration sensor (not shown) disposed inside or outside the camera module **180**. According to an embodiment, the image stabilizer **240** may be implemented, for example, as an optical image stabilizer. The memory **250** may store, at least temporarily, at least part of an image obtained via the image sensor **230** for a subsequent image processing task. For example, if image capturing is delayed due to shutter lag or multiple images are quickly captured, a raw image obtained (e.g., a Bayer-patterned image, a high-resolution image) may be stored in the memory **250**, and its corresponding copy image (e.g., a low-resolution image) may be previewed via the display module **160**. Thereafter, if a specified condition is met (e.g., by a user's input or system command), at least part of the raw image stored in the memory **250** may be obtained and processed, for example, by the image signal processor **260**. According to an embodiment, the memory **250** may be configured as at least part of the memory **130** or as a separate memory that is operated independently from the memory **130**.

[0058] The image signal processor **260** may include various circuitry and perform one or more image processing with respect to an image obtained via the image sensor **230** or an image stored in the memory **250**. The one or more image processing may include, for example, depth map generation, three-dimensional (3D) modeling, panorama generation, feature point extraction, image synthesizing, or image compensation (e.g., noise reduction, resolution adjustment, brightness adjustment, blurring, sharpening, or softening). Additionally or alternatively, the image signal

processor **260** may perform control (e.g., exposure time control or read-out timing control) with respect to at least one (e.g., the image sensor **230**) of the components included in the camera module **180**. An image processed by the image signal processor **260** may be stored back in the memory **250** for further processing, or may be provided to an external component (e.g., the memory **130**, the display module **160**, the electronic device **102**, the electronic device **104**, or the server **108**) outside the camera module **180**. According to an embodiment, the image signal processor **260** may be configured as at least part of the processor **120**, or as a separate processor that is operated independently from the processor **120**. If the image signal processor **260** is configured as a separate processor from the processor **120**, at least one image processed by the image signal processor **260** may be displayed, by the processor **120**, via the display module **160** as it is or after being further processed.
[0059] According to an embodiment, the electronic device **101** may include a plurality of camera modules **180** having different attributes or functions. In such a case, at least one of the plurality of camera modules **180** may form, for example, a wide-angle camera and at least another of the plurality of camera modules **180** may form a telephoto camera. Similarly, at least one of the plurality of camera modules **180** may form, for example, a front camera and at least another of the plurality of camera modules **180** may form a rear camera.
[0060] FIG. **3** is an exploded perspective view of an example electronic device **101** according to various embodiments. FIG. **4** is a partial exploded perspective view of the electronic device **101** of FIG. **3** according to various embodiments.
[0061] Referring to FIGS. **3** and **4**, the electronic device **101** according to an embodiment may include a housing **340**, a printed circuit board **310**, a camera module (e.g., the camera module **180** of FIG. **2**), a camera housing **320**, and a bracket **330**. For example, the electronic device **101** may be a mobile terminal including a front camera and/or a rear camera, but is not limited thereto. The electronic device **101** may include one or more camera modules. According to an embodiment, the camera module **180** may include a camera **181** including a lens **210**.
[0062] According to an embodiment, the housing **340** may form an exterior surface of the electronic device **101**. For example, the housing **340** may include a support **344** configured to support components of the electronic device **101**, a first plate **341** surrounding the support **344** and/or a second plate **342** coupled to the first plate **341** and disposed on the support **344**. For example, the first plate **341** may be referred to as a side member forming side surfaces of the electronic device **101**. For example, the support **344** may be referred to as a supporting portion extending from the first plate **341** toward an inside of the housing **340**. For example, the first plate **341** and the support **344** may be integrated, but are not limited thereto. The camera module **180** and/or the printed circuit board **310** may be disposed on the support **344**. The second plate **342** may face a first direction D**1**. The second plate **342** may include a camera decoration **342***a* for covering the camera module **180**. Although not illustrated, the housing **340** may further include a third plate facing the second plate **342**. For example, the third plate may face a fourth direction D**4** opposite to the first direction D**1**.
[0063] According to an embodiment, the printed circuit board **310** may be disposed on a surface of the support **344** facing the first direction D**1**. For example, the printed circuit board **310** and the camera **181** may be disposed side by side on the support **344**. For example, the printed circuit board **310** and the camera **181** may be electrically connected through electrical wires. The printed circuit board **310** may include a plurality of conductive layers and a plurality of non-conductive layers alternately laminated with the plurality of conductive layers. For example, the printed circuit board **310** may include a ground portion for grounding electronic components. The printed circuit board **310** may provide an electrical connection between various electronic components of the electronic device **101**, using wires and conductive vias formed on the conductive layer. For example, the camera **181** may be electrically connected to a processor (e.g., the processor **120** of FIG. **1**) through the printed circuit board **310**.

[0064] According to an embodiment, the camera **181** may be disposed on the support **344**. The camera **181** may include a lens **210** facing the first direction D**1**. FIG. **3** illustrates that the camera **181** faces the second plate **342** of the electronic device **101**, but the disclosure is not limited thereto. For example, the first direction D**1** may be a direction in which the second plate **342** faces or an opposite direction to a direction in which the second plate **342** faces. The camera **181** may include one or more lenses. For example, the lens **210** may collect light emitted from a subject that is a target of an image capture. For example, the lens **210** may include a wide-angle lens or a telephoto lens, but is not limited thereto. According to an embodiment, the electronic device **101** may include a plurality of cameras disposed at different positions. Although not illustrated, the camera **181** may further include an actuator for driving the lens **210** and/or an image sensor (e.g., the image sensor **230** of FIG. **2**).

[0065] According to an embodiment, the camera housing **320** may surround at least a portion of the camera **181**. For example, the camera housing **320** may be referred to as a shield can. Referring to FIG. **4**, the camera housing **320** may include a first surface **321** facing the first direction D**1** and a second surface **322** substantially perpendicular to the first surface **321**. The camera **181** may be disposed within a space surrounded by the first surface **321** and the second surface **322**. The camera housing **320** may include an opening **323** for a lens **210** exposed toward an outside of the camera housing **320** within the first surface **321**. A portion of the lens **210** may pass through the opening **323**. For example, the lens **210** may be exposed from an inside of the camera housing **320** through the opening **323** to an outside of the first surface **321** of the camera housing **320** facing the first direction D**1**. The lens **210** may be configured to collect light emitted from a subject by being exposed to the outside of the camera housing **320** through the opening **323**. According to an embodiment, the camera housing **320** may protect the camera **181** accommodated therein. The camera housing **320** may shield electromagnetic waves emitted from other electronic components (e.g., an antenna module) of the electronic device **101** from reaching the camera **181**. The camera housing **320** may be configured to shield electromagnetic waves emitted from the camera **181**.

[0066] According to an embodiment, the bracket **330** may at least partially surround the camera housing **320**. The bracket **330** may be fixed to the support **344** and/or the printed circuit board **310** to fix a position of the camera **181** at a designated position within the housing **340**. For example, the bracket **330** may include an accommodating space **330***a* for surrounding at least a portion of the camera **181**. The camera housing **320** may be disposed within the accommodating space **330***a*.

[0067] As various electronic components (e.g., speaker, microphone, or antenna) are disposed in the electronic device **101**, electromagnetic compatibility (EMC) of the camera **181** may be required. For example, in the housing **340**, an antenna module (e.g., the antenna module **197** of FIG. **1**) may be adjacent to the camera **181**. When a transmission signal transmitted from an antenna module to an external electronic device is coupled to the camera **181** adjacent to the antenna module, malfunction of some components (e.g., image sensor and/or actuator) of the camera **181** may occur. For example, when electromagnetic waves emitted during operations of some components (e.g., image sensor and/or actuator) of the camera **181** is induced to an antenna module adjacent to the camera **181**, performance (e.g., sensitivity) of the antenna module may be degraded. According to an embodiment, the camera housing **320** surrounding at least a portion of the camera **181** may be required to be electrically grounded to shield electromagnetic waves emitted from the camera **181** and shield electromagnetic waves emitted from other electronic components from reaching the camera **181**. Enhancing the grounding performance of the camera housing **320** may improve electromagnetic interference (EMI) and/or electromagnetic susceptibility (EMS). When the camera housing **320** is directly connected to a ground portion of the printed circuit board **310**, a separate connecting member (e.g., a connector including, for example, a c-clip or conductive tape) may be required for an electrical connection between the camera housing **320** and the ground portion. For example, when the camera housing **320** is electrically connected to the ground portion through the separate connecting member, an internal short may occur due to

separation of the connecting member.
[0068] According to an embodiment, the camera housing **320** surrounding at least a portion of the camera **181** may be grounded for EMC of the camera **181** via the bracket **330**. For example, the bracket **330** may be connected to a ground portion of the printed circuit board **310** and/or a ground region of the support **344**. The camera housing **320** at least partially surrounded by the bracket **330** may be connected to the bracket **330**. According to an embodiment, the camera housing **320** may include a bridge (e.g., the bridge **324** of FIG. **5**A) for being connected to the bracket **330**. The bridge **324** may be implemented as a portion of the camera housing **320**, rather than as a separate connecting member independent of the camera housing **320**. The camera housing **320** electrically connected to the ground region may shield electromagnetic waves reaching the camera **181** from other electronic components, and/or electromagnetic waves emitted from the camera **181**. As the camera housing **320** is grounded, electromagnetic interference between the camera **181** and other electronic components (e.g., speaker, microphone, or antenna) within the electronic device **101** may be reduced. According to an embodiment, the electronic device **101** may reduce malfunctions and/or degradation of signal quality due to noise, by reducing electromagnetic interference.
[0069] FIG. **5**A is a perspective view illustrating an example camera housing **320** according to various embodiments. FIG. **5**B is a diagram illustrating an example process of forming an example camera housing **320** according to various embodiments. FIG. **6** is a cross-sectional view of an example electronic device **101** taken along line A-A′ of FIG. **3** according to various embodiments.
[0070] Referring to FIG. **5**A, the camera housing **320** may include a first surface **321** and a second surface **322** substantially perpendicular to the first surface **321**. The second surface **322** may extend substantially perpendicularly with respect to the first surface **321** from a periphery **321***a* of the first surface **321**. An opening **323** may be formed in the first surface **321**. A camera (e.g., the camera **181** of FIG. **4**) may be disposed within a space surrounded by the first surface **321** and the second surface **322**. The space surrounded by the first surface **321** and the second surface **322** may be exposed to the outside of the camera housing **320** through the opening **323**. At least a portion of a lens (e.g., the lens **210** of FIG. **4**) of the camera **181** may be exposed to the outside of the camera housing **320** through the opening **323**.
[0071] According to an embodiment, the camera housing **320** may include at least one bridge **324**. Referring to FIG. **5**B, the bridge **324** may be formed during a process of forming the opening **323** of the first surface **321**. States **501**, **502**, and **503** of FIG. **5**B schematically illustrate the first surface **321** of the camera housing **320** in the process of forming the camera housing **320** including the bridge **324**.
[0072] For example, the state **501** indicates a top view of the camera housing **320** before the opening **323** is formed. Before the opening **323** is formed, the first surface **321** may close the internal space.
[0073] The state **502** indicates a top view of the camera housing **320** in a state that an opening **323** is formed by processing a portion of the first surface **321**. For example, the opening **323** may be formed by a press method of punching a portion of the first surface **321**, but is not limited thereto. In the process of forming the opening **323**, the bridge **324** extending from a portion of a periphery **323***a* of the opening **323** within the first surface **321** toward a center of the opening **323** may be formed. For example, the opening **323** may be formed by leaving the bridge **324** in the first surface **321** and removing a remaining portion.
[0074] According to an embodiment, the bridge **324** may be one or more. For example, the camera housing **320** may include a pluralidty of bridges **325***a*, **325***b*, **325***c*, and **325***d* formed along the periphery **323***a* of the opening **323**. The plurality of bridges **325***a*, **325***b*, **325***c*, and **325***d* may be symmetrical to each other. For example, the camera housing **320** may include four bridges **325***a*, **325***b*, **325***c*, and **325***d* that are symmetrical to each other with respect to the center of the opening **323**. For example, the four bridges **325***a*, **325***b*, **325***c*, and **325***d* may be disposed to be orthogonal to each other at 90 degrees. For example, a first bridge **325***a* and a third bridge **325***c* may face each

other. For example, a second bridge **325***b* and a fourth bridge **325***d* may face each other. However, the disclosure is not limited thereto. When the plurality of bridges **325***a*, **325***b*, **325***c*, and **325***d* are disposed symmetrically, the area for grounding the camera housing **320** may increase, so the grounding effect of the camera housing **320** may be improved.

[0075] The state **503** indicates a top view of the camera housing **320** in a state that the bridge **324** is folded toward the outside of the camera housing **320**. In the state **502**, the bridge **324** may extend from the periphery **323***a* of the opening **323** toward the center of the opening **323**. A portion of the bridge **324** may be folded toward the outside of the camera housing **320** so that the bridge **324** is contacted with a bracket (e.g., the bracket **330** of FIG. **6**) at least partially surrounding the camera housing **320**. The folded portion of the bridge **324** may be referred to as a bending portion (e.g., the bending portion **324***a* of FIG. **6**).

[0076] According to an embodiment, the bridge **324**, which was extended toward the center of the opening **323**, may be folded to face the outside of the camera housing **320**. The bridge **324** may be extended toward the bracket **330** through the bending portion **324***a* and may be contacted with the bracket **330**. The camera housing **320** may be electrically connected to a ground portion that is electrically connected to the bracket **330**, through the bridge **324** contacted with the bracket **330**.

[0077] Referring to FIG. **6**, the bracket **330** may include a third surface **331** faced away from the second surface **322**. The third surface **331** being faced away from the second surface **322** may refer to a structure in which a direction in which the third surface **331** faces is opposite to a direction in which the second surface **322** faces, and the third surface **331** is spaced apart from the second surface **322**. The bracket **330** may be electrically connected to a ground portion of the printed circuit board **310**. The bridge **324** may be folded from a portion of the periphery **323***a* of the opening **323** formed in the first surface **321** toward the periphery **321***a* of the first surface **321**, thereby contacting with the third surface **331**.

[0078] According to an embodiment, the bridge **324** may extend from a portion of the periphery **323***a* of the opening **323** to the third surface **331** faced away from the second surface **322**. The third surface **331** may be referred to as an inner surface of the bracket **330** facing the camera housing **320**. For example, a gap g may be formed between the second surface **322** of the camera housing **320** and the third surface **331** of the bracket **330**. The bridge **324** may extend to the third surface **331** through the gap g between the second surface **322** and the third surface **331**. As the bridge **324** is contacted with the third surface **331** of the bracket **330**, the camera housing **320** and the bracket **330** may be electrically connected. Since the bracket **330** is electrically connected to the ground portion of the printed circuit board **310**, the camera housing **320** may be electrically connected to the ground portion of the printed circuit board **310** through the bracket **330** contacted with the bridge **324**.

[0079] According to an embodiment, the bridge **324** may include a bending portion **324***a*. The bending portion **324***a* may be formed at an end portion of the bridge **324**. The end potion of the bridge **324** may be referred to as a portion that contacts a portion of the periphery **323***a* of the opening **323**. If the bending portion **324***a* protrudes toward the opening **323** when folded, it may affect the lens **210** passing through the opening **323**. In order to prevent and/or reduce the bending portion **324***a* from interfering with the lens **210**, the first surface **321** of the camera housing **320** may include a cut portion (e.g., the cut portion **321***b* of FIG. **5**A) cut out from a portion of the periphery **323***a* of the opening **323**, in which the bridge **324** extends, toward the periphery **321***a* of the first surface **321**. For example, the cut portion **321***b* may extend from a portion of the periphery **323***a* of the opening **323** toward the periphery **321***a* of the first surface **321** with a width corresponding to a width of the bridge **324**. A length of the cut portion **321***b* may be less than or equal to a length between the periphery **323***a* of the opening **323** and the periphery **321***a* of the first surface **321**. When the bending portion **324***a* is folded, the cut portion **321***b* may form the bending portion **324***a*, so the bending portion **324***a* may be disposed on the outside of the opening **323**. According to an embodiment, since the bending portion **324***a* is disposed on the outside of the

opening **323**, interference between the bending portion **324***a* and the lens **210** may be reduced.
[0080] According to an embodiment, the bending portion **324***a* may be bent from a portion of the periphery **323***a* of the opening **323** in a third direction D**3** that is opposite to a second direction D**2** toward the lens **210** passing through the opening **323**. The bridge **324** may extend toward the bracket **330** disposed on the outside of the camera housing **320**, through the bending portion **324***a*. According to an embodiment, the bridge **324** may include the bending portion **324***a*, a first portion **324-1**, and a second portion **324-2**. The first portion **324-1** may extend from the bending portion **324***a* in the third direction D**3**. The second portion **324-2** may extend from the first portion **324-1** with an inclination with respect to the first portion **324-1**. For example, the second portion **324-2** may extend in a direction between the third direction D**3** and a fourth direction D**4** opposite to the first direction D**1**. For example, the second portion **324-2** may extend to a gap g between the second surface **322** and the third surface **331** and may contact the third surface **331**. The camera housing **320** may be electrically connected to a ground portion, via the bridge **324** contacted with the third surface **331** of the bracket **330** electrically connected to the ground portion. The camera housing **320** electrically connected to the ground portion may be configured to shield electromagnetic waves reaching the camera **181** from other electronic components, and/or electromagnetic waves emitted from the camera **181**.
[0081] According to an embodiment, the bridge **324** may be disposed between the camera housing **320** and the bracket **330** to fix the camera housing **320** within the bracket **330**. For example, the bridge **324** may have elasticity. For example, the camera housing **320** may include an elastic material (e.g., stainless steel). The bridge **324** extending from the camera housing **320** may include the same material as the camera housing **320**. For example, the bridge **324** may have a width capable of providing elasticity. The camera housing **320** within the bracket **330** may be coupled to the bracket **330** by the elasticity of the bridge **324**. When the camera housing **320** is coupled into the bracket **330**, the bridge **324** may be compressed to be less than or equal to a gap g between the bracket **330** and the camera housing **320**. After the camera housing **320** is inserted within the bracket **330**, the camera housing **320** may be supported by the restoring force of the bridge **324**. The bridge **324** may be configured to support the camera housing **320** with respect to the bracket **330**.
[0082] According to an embodiment, since the camera housing **320** may be coupled to the bracket **330** by utilizing the elasticity of the bridge **324**, the stress applied to the camera housing **320** may be reduced. For example, when the camera housing **320** is inserted into the bracket **330** through at least one protrusion formed on a side surface of the camera housing **320** (e.g., force-fitting), the camera **181** within the camera housing **320** may continuously receive stress applied from the protrusion. Since the camera **181** has at least one circuit that is vulnerable to damage, it may be damaged by continuously applied stress. In order to protect the camera **181** having a sensitive circuit, the bridge **324** may be configured to reduce the stress due to coupling by supporting the camera housing **320** using elasticity. According to an embodiment, since the bridge **324** extends from the first surface **321** of the camera housing **320**, the stress due to the elastic force of the bridge **324** may be transferred to the first surface **321** of the camera housing **320**. Since the stress due to the elastic force of the bridge **324** may be transferred to the first surface **321** when the bridge **324** supports the camera housing **320** with respect to the bracket **330**, the stress applied to the camera **181** disposed inside the camera housing **320** may be reduced.
[0083] According to an embodiment, the bracket **330** may include a groove **332** formed within the third surface **331**. The bridge **324** may include an inserting portion **324***b* inserted within the groove **332**. For example, the groove **332** may be recessed into the interior of the bracket **330** from the third surface **331**. A shape of the inserting portion **324***b* may correspond to a shape of the groove **332**. When the camera housing **320** is coupled into the bracket **330**, a position of the inserting portion **324***b* may correspond to a position of the groove **332**. The camera housing **320** may be fixed within the bracket **330** through the inserting portion **324***b* inserted within the groove **332**. The

inserting portion **324***b* may prevent and/or suppress the camera housing **320** from being separated from the bracket **330**. The contact area of the bridge **324** and the bracket **330** may be increased through the inserting portion **324***b* and the groove **332** that are contacted with each other. As the contact area of the bridge **324** and the bracket **330** is increased, an electrical connection between the camera housing **320** and the bracket **330** may be improved, so the grounding effect of the camera housing **320** may be improved.

[0084] FIGS. **7**A and **7**B are cross-sectional views illustrating the camera housing **320** illustrated in FIG. **6** being coupled to a bracket **330** according to various embodiments.

[0085] According to an embodiment, the shape of the bridge **324** may be different according to a coupling direction of the camera housing **320** and the bracket **330**.

[0086] Referring to FIG. **7**A, the camera housing **320** including the bridge **324** including the first portion **324-1** and the second portion **324-2** described above may be coupled to the bracket **330** in the first direction D**1**. In the gap g between the second surface **322** and the third surface **331**, the second portion **324-2** may extend in a direction between the third direction D**3** and the fourth direction D**4**. Since the second portion **324-2** has an incline with respect to the first portion **324-1**, when an end **324-2***a* of the second portion **324-2** connected to the first portion **324-1** is inserted into the bracket **330** before another end **324-2***b* of the second portion **324-2**, the camera housing **320** may be naturally coupled into the bracket **330**. After the end **324-2***a* of the second portion **324-2** is first inserted into the bracket **330**, when the camera housing **320** is pushed into the bracket **330** in the first direction D**1**, the second portion **324-2** may be naturally compressed and inserted into the bracket **330**. The other end **324-2***b* of the second portion **324-2** may be contacted with the third surface **331** of the bracket **330**. When the groove **332** is formed in the bracket **330**, at least a portion of the second portion **324-2** may be inserted into the groove **332**. After the second portion **324-2** is fully inserted into the bracket **330**, the camera housing **320** may be supported relative to the bracket **330** by the elastic force of the bridge **324**. The camera housing **320** of the above structure may be naturally inserted into the bracket **330** when it is coupled into the bracket **330** in the first direction D**1**.

[0087] Referring to FIG. **7**B, when the camera housing **320** including the bridge **324** including the first portion **324-1** and the second portion **324-2** is coupled to the bracket **330** in the fourth direction D**4**, the other end **324-2***b* of the second portion **324-2** may be inserted into the bracket **330** before the end **324-2***a* of the second portion **324-2**. In order to form the elastic force of the bridge **324** inside the bracket **330**, the position of the bridge **324** may be designed to be tightly coupled to the internal space of the bracket **330**. When the other end **324-2***b* of the second portion **324-2** is inserted into the bracket **330** before the first end **324-2***a* of the second portion **324-2**, the other end **324-2***b* may be positioned on the outside of the opening **323** (e.g., in the third direction D**3**). When the camera housing **320** of the above structure is coupled in the fourth direction D**4**, the bridge **324** may be damaged by being caught on the opening **323**, or insertion of the camera housing **320** may be difficult.

[0088] FIG. **8**A is a cross-sectional view of an example electronic device **101** taken along line A-A′ of FIG. **3** according to various embodiments. FIGS. **8**B and **8**C are cross-sectional views illustrating the camera housing **320** illustrated in FIG. **8**A being coupled to a bracket **330** according to various embodiments.

[0089] Referring to FIG. **8**A, the bridge **324** may include a bending portion **324***a*, a first portion **324-1**, a second portion **324-2**, and a third portion **324-3**. A structure of the bridge **324** illustrated in FIG. **8**A may be a structure suitable for the camera housing **320** to be coupled to the bracket **330** in the fourth direction D**4**.

[0090] According to an embodiment, the second portion **324-2** may extend from the first portion **324-1** to a point between the second surface **322** and the third surface **331**. The second portion **324-2** may extend to a gap g between the second surface **322** and the third surface **331** without contacting the third surface **331** of the bracket **330**. The third portion **324-3** may extend from the second portion **324-2** in a direction between the first direction D**1** and the third direction D**3**, and may contact the third surface **331**. For example, the third portion **324-3** may extend from an end **324-3***a* connected to the second portion **324-2** to another end **324-3***b*. The third portion **324-3** may contact the third surface **331** of the bracket **330**, or, when a groove **332** is formed in the bracket **330**, at least a portion of the third portion **324-3** may be inserted into the groove **332** of the bracket **330**. An angle between the second portion **324-2** and the third portion **324-3** may be an acute angle (e.g., the angle A of FIG. **8**A). Since the third portion **324-3** is bent relative to the second portion **324-2**, the bridge **324** may include a bending portion between the bent portion **324***a* and the second portion **324-2** and the third portion **324-3**.

[0091] Referring to FIG. **8**B, the camera housing **320** including the bridge **324** including the bending portion **324***a*, the first portion **324-1**, the second portion **324-2**, and the third portion **324-3** may be coupled to the bracket **330** in the fourth direction D**4**. Since the third portion **324-3** extends in a direction between the first direction D**1** and the third direction D**3**, when the end **324-3***a* of the third portion **324-3** connected to the second portion **324-2** is inserted into the bracket **330** before the other end **324-3***b* of the third portion **324-3**, the camera housing **320** may be naturally coupled into the bracket **330**. For example, before the camera housing **320** is coupled into the bracket **330**, the end **324-3***a* of the third portion **324-3** may be positioned inward of the third surface **331** of the bracket **330** (e.g., in the second direction D**2**). When the end **324-3***a* of the third portion **324-3** is first inserted into the bracket **330** and then the camera housing **320** is pushed into the bracket **330** in the fourth direction D**4**, the second portion **324-2** and the third portion **324-3** may be naturally compressed and inserted into the bracket **330**. For example, the other end **324-3***b* of the third portion **324-3** may be contacted with the third surface **331** of the bracket **330**. When a groove **332** is formed in the bracket **330**, at least a portion of the third portion **324-3** may be inserted into the groove **332**. After the second portion **324-2** and the third portion **324-3** are fully inserted into the bracket **330**, the camera housing **320** may be supported by the bracket **330** by the elastic force of the bridge **324**. The camera housing **320** of the above structure may be naturally inserted into the bracket **330** when it is coupled into the bracket **330** in the fourth direction D**4**.

[0092] Referring to FIG. **8**C, when the camera housing **320** including the bridge **324** including the bending portion **324***a*, the first portion **324-1**, the second portion **324-2**, and the third portion **324-3** is coupled to the bracket **330** in the first direction D**1**, the other end **324-3***b* of the third portion **324-3** protruding toward the outside of the camera housing **320** may be positioned on the outside of the third surface **331** of the bracket **330** (e.g., the third direction D**3**). When the camera housing **320** is pressed in the first direction D**1** in a state that a portion of the third portion **324-3** is positioned on the outside of the third surface **331**, the bridge **324** may be damaged by being caught on the opening **323**, or insertion of the camera housing **320** may be difficult.

[0093] According to an embodiment, a shape of the bridge **324** may vary according to a coupling direction of the camera housing **320** and the bracket **330**. For example, when the camera housing **320** is required to be coupled to the bracket **330** in the first direction D**1**, the bridge **324** may include the first portion **324-1** and the second portion **324-2**. For example, when the camera housing **320** is required to be coupled to the bracket **330** in the fourth direction D**4**, the bridge **324** may include the first portion **324-1**, the second portion **324-2**, and the third portion **324-3**. According to an embodiment, the camera housing **320** is not limited to a specific coupling direction and may have a bridge **324** with a suitable structure according to the coupling direction.

[0094] FIG. **9** is a perspective view illustrating an example camera housing **320** according to various embodiments. FIGS. **10**A and **10**B are cross-sectional views of an example electronic device **101** including the camera housing **320** of FIG. **9** taken along line A-A′ of FIG. **3** according to various embodiments.

[0095] Referring to FIGS. **9**, **10**A, and **10**B, the bridge **324** may extend from the second surface **322** of the camera housing **320**. For example, the second surface **322** may extend substantially perpendicularly to the first surface **321**. A direction in which the second surface **322** extends may be a fourth direction D**4** opposite to a first direction D**1** in which the first surface **321** faces. The bridge **324** may extend from a portion of the other end **322***b* of the second surface **322** opposite to the end **322***a* of the second surface **322** connected to the first surface **321**, in a direction between the first direction D**1** and the third direction D**3**. The other end **322***b* may be referred to as an end of the fourth direction D**4** of the second surface **322**.

[0096] According to an embodiment, the bridge **324** may extend from a portion of the other end **322***b* of the second surface **322** to the third surface **331** of the bracket **330**. The bridge **324** may connect the camera housing **320** and the bracket **330**. The camera housing **320** may be electrically connected to a ground portion of the printed circuit board **310** through the bracket **330**. The bridge **324** formed on the portion of the other end **322***b* of the second surface **322** may support the camera housing **320** relative to the bracket **330** in the gap g between the second surface **322** and the third surface **331**. When the camera housing **320** is inserted into the bracket **330**, the bridge **324** having elasticity may be inserted in a compressed state. In the bracket **330**, the bridge **324** may be coupled, by restoring force, between the third surface **331** of the bracket **330** and the second surface **322** of the camera housing **320**.

[0097] According to an embodiment, the force by which the bridge **324** supports the camera housing **320** may be provided to an end of the camera housing **320**. For example, the stress applied to the camera housing **320** by the elastic force of the bridge **324** may be transferred to the other end **322***b* of the second surface **322** of the camera housing **320**. When the bridge **324** supports the camera housing **320** relative to the bracket **330**, the stress due to the elastic force of the bridge **324** may be transferred to the other end **322***b* of the second surface **322**, so the stress applied to the camera **181** disposed inside the camera housing **320** may be reduced. According to an embodiment, since at least one circuit inside the camera **181** may receive less stress due to the elastic force of the bridge **324**, damage to the at least one circuit may be reduced.

[0098] According to an embodiment, a shape of the bridge **324** may vary according to a coupling direction of the camera housing **320** and the bracket **330**. Referring to FIG. **10**A, the bridge **324** may include the first portion **324-1**. The first portion **324-1** may extend from a portion of the other end **322***b* of the second surface **322** in a direction between the first direction D**1** and the third direction D**3**. The first portion **324-1** may extend to the gap g between the second surface **322** and the third surface **331** and may contact the third surface **331**. When a groove **332** is formed in the bracket **330**, at least a portion of the first portion **324-1** may be inserted into the groove **332**.

[0099] The camera housing **320** of the structure illustrated in FIG. **10**A may be coupled to the bracket **330** in the fourth direction D**4**. When the camera housing **320** is coupled to the bracket **330** in the fourth direction D**4**, an end portion **324-1***a* of the first portion **324-1** connected to the second surface **322** may be inserted into the bracket **330** earlier than another end portion **324-1***b* of the first portion **324-1** opposite to the end portion **324-1***a*. Since the other end portion **324-1***b* is positioned further outward from the camera housing **320** than the end portion **324-1***a*, when the camera housing **320** is inserted into the bracket **330** in the fourth direction D**4**, the first portion **324-1** may be naturally compressed toward the camera housing **320** by the bracket **330**. For example, the other end portion **324-1***b* may be inserted into the groove **332** of the bracket **330**. In a state that the camera housing **320** is fully inserted into the bracket **330**, the bridge **324** may support the camera housing **320** relative to the bracket **330**.

[0100] Referring to FIG. **10**B, the bridge **324** may include a first portion **324-1** and a second portion **324-2**. The first portion **324-1** may extend from a portion of the other end portion **322***b* of the second surface **322** in a direction between the first direction D**1** and the third direction D**3**. The second portion **324-2** may extend from the first portion **324-1** in a direction between the third direction D**3** and the fourth direction D**4** and may contact the third surface **331**. When the groove **332** is formed in the bracket **330**, at least a portion of the second portion **324-2** may be inserted into the groove **332**. An angle between the first portion **324-1** and the second portion **324-2** may be an acute angle (e.g., the angle B of FIG. **10**B). Since the second portion **324-2** is bent relative to the first portion **324-1**, the bridge **324** may include a bent portion between the first portion **324-1** and the second portion **324-2**.

[0101] The camera housing **320** of the structure illustrated in FIG. **10***b* may be coupled to the bracket **330** in the first direction D**1**. When the camera housing **320** is coupled to the bracket **330** in the first direction D**1**, an end portion **324-2***a* of the second portion **324-2** connected to the first portion **324-1** may be inserted into the bracket **330** earlier than another end portion **324-2***b* of the second portion **324-2** spaced from the first portion **324-1**. Since the other end portion **324-2***b* is positioned further outward from the camera housing **320** than the end portion **324-2***a*, when the camera housing **320** is inserted into the bracket **330** in the first direction D**1**, the bridge **324** may be naturally compressed toward the camera housing **320** by the bracket **330**. For example, the other end portion **324-2***b* may be inserted into the groove **332** of the bracket **330**. In a state that the camera housing **320** is fully inserted into the bracket **330**, the bridge **324** may support the camera housing **320** relative to the bracket **330**.

[0102] According to an embodiment, the camera housing **320** having the bridge **324** extending from a portion of the other end portion **322***b* of the second surface **322** may be electrically connected to a ground portion of a printed circuit board **310** through the bracket **330**. The camera housing **320** electrically connected to the ground portion may shield electromagnetic waves reaching the camera **181** from other electronic components and/or electromagnetic waves emitted from the camera **181** within the camera housing **320**. Through the bridge **324**, the camera housing **320** may be fixed within the bracket **330**. Since the bridge **324** is formed on a portion of the other end portion **322***b* of the second surface **322**, the stress applied to the camera housing **320** may not be transferred to the camera **181** but may be transferred to the other end portion **322***b* of the second surface **322**. According to an embodiment, damage to the camera **181** due to the stress may be reduced. According to an embodiment, a shape of the bridge **324** may be different according to a coupling direction of the camera **181**.

[0103] FIG. **11**A is a perspective view illustrating an example camera housing **320** being connected to a printed circuit board **310** according to various embodiments. FIG. **11**B is a perspective view illustrating an example camera housing **320** being connected to a support **344**. FIG. **11**C is a diagram illustrating an example process of forming an example camera housing **320** according to various embodiments.

[0104] Referring to FIG. **11**A, the camera housing **320**, which is disposed within the bracket **330**, may be directly connected to the printed circuit board **310**. According to an embodiment, the bridge **324** may pass through the bracket **330** and extend to a ground portion of the printed circuit board **310**. When the bridge **324** is connected to the ground portion, the camera housing **320** may be electrically connected to the ground portion through the bridge **324**.

[0105] According to an embodiment, the bracket **330** may include a through hole **333**. The through hole **333** may be formed on a side surface of the bracket **330**. For example, the through hole **333** may be formed on a side surface of the bracket **330** adjacent to the printed circuit board **310**. The bridge **324** extending from the first surface **321** of the camera housing **320** may pass through the through hole **333**. For example, the bridge **324** may have a length that passes between the camera housing **320** and the bracket **330** and extends through the through-hole **333**. The bridge **324** exposed through the through hole **333** may be connected to the ground portion of the printed circuit board **310**. For example, the bridge **324** may be connected to the ground portion through a conductive connecting member (e.g., c-clip) for electrical connection. Since the bridge **324** is connected to the ground portion, the camera housing **320** may be directly grounded, without passing through the bracket **330**. The grounded camera housing **320** may shield electromagnetic waves reaching the camera **181** from other electronic components and/or electromagnetic waves emitted from the camera **181** disposed inside the camera housing **320**.

[0106] Referring to FIG. **11**B, the camera housing **320** disposed within the bracket **330** may be directly connected to a ground region of the support **344**. According to an embodiment, the bridge

324 may extend through the bracket **330** to the ground region of the support **344**. When the bridge **324** is connected to the ground region, the camera housing **320** may be electrically connected to the ground portion through the bridge **324**.

[0107] According to an embodiment, the bracket **330** may include a through hole **333**. The through hole **333** may be formed on a side surface of the bracket **330**. For example, the through hole **333** may be formed on a side surface of the bracket **330** adjacent to the ground region of the support **344**. The bridge **324** extending from the first surface **321** of the camera housing **320** may pass through the through hole **333**. For example, the bridge **324** may have a length that passes between the camera housing **320** and the bracket **330** and extends through the through hole **333**. The bridge **324** exposed through the through hole **333** may be connected to the ground region of the support **344**.

[0108] According to an embodiment, the bracket **330** may include a protruding portion **334** into which a screw **350** is inserted. For example, the protruding portion **334** may be contacted with the through hole **333**. The screw **350** may penetrate the bridge **324** and the protruding portion **334**, and may be inserted into the grounding region. Since the screw **350** penetrates the bridge **324** and the protruding portion **334**, the camera housing **320** may be electrically connected to the grounding region. For example, the screw **350** may include a conductive material (e.g., metal), and the bridge **324** may be electrically connected to the ground region through the screw **350**. Since the bridge **324** is connected to the ground region through the screw **350**, the camera housing **320** may be directly grounded without passing through the bracket **330**. The grounded camera housing **320** may shield electromagnetic waves reaching the camera **181** from other electronic components and/or electromagnetic waves emitted from a camera (e.g., the camera **181** of FIG. **6**) disposed inside the camera housing **320**. The screw **350** may fix the bracket **330** to the support **344**. Since the screw **350** may penetrate the protruding portion **334** and be inserted into the support **344**, the bracket **330** may be fixed at a designated position on the support **344**.

[0109] According to an embodiment, in order for the bridge **324** to pass through the bracket **330** and be connected to the printed circuit board **310** and/or the support **344**, it may be required to secure a length of the bridge **324**. Since the length of the bridge **324** may be limited by the opening **323**, in order to secure the length of the bridge **324**, the bridge **324** may be formed to have the longest length within the opening **323**.

[0110] **1101** of FIG. **11**C illustrates a first surface **321** when the camera housing **320** is viewed from above. **1102** of FIG. **11**C illustrates a side surface of the camera housing **320**. Referring to **1101** of FIG. **11**C, the bridge **324** may extend from a portion of the periphery **323***a* of the opening **323** toward a center of the opening **323**. In order for the bridge **324** to be formed to be the longest in the opening **323**, the bridge **324** may be formed between two points that are farthest apart from each other in the periphery **323***a* of the opening **323**. For example, a portion of the periphery within the first surface **321** where the bridge **324** is formed may be positioned at a corner C of the opening **323**. For example, the bridge **324** may be formed between two points diagonally crossing the opening **323**, but is not limited thereto. Referring to **1102** of FIG. **11**C, the length of the bridge **324** may be formed relatively long. The elongated bridge **324** may have a length that may pass through a through-hole (e.g., the through-hole **333** of FIG. **11**A) of a bracket (e.g., the bracket **330** of FIG. **11**A) and extend to the printed circuit board **310** and/or the support **344**. According to an embodiment, when the camera housing **320** is directly connected to the printed circuit board **310** and/or the support **344**, the grounding effect of the camera housing **320** may be enhanced. Since the camera housing **320** may shield electromagnetic waves emitted from the camera **181**, the influence of the camera **181** on other electronic components may be reduced. Since the camera housing **320** may shield electromagnetic waves reaching the camera **181** from other electronic components, malfunctions of the camera **181** may be reduced.

[0111] FIG. **12**A is a perspective view illustrating the camera housing **320** illustrated in FIG. **11**A or **11**B being coupled to a bracket **330** in a first direction D**1** according to various embodiments. FIG.

12B is a perspective view illustrating the camera housing **320** illustrated in FIG. **11**A or **11**B being coupled to a bracket **330** in a direction opposite to the first direction D**1** according to various embodiments. FIG. **12**C is a perspective view illustrating the camera housing **320** illustrated in FIG. **10**B being coupled to a bracket **330** in a fourth direction D**4** opposite to the first direction D**1** according to various embodiments.

[0112] According to an embodiment, a position of the through hole **333** of the bracket **330** may be different based on a direction in which the camera housing **320** is coupled to the bracket **330** or a position in which the bridge **324** is formed.

[0113] Referring to FIG. **12**A, the camera housing **320** may be inserted into the bracket **330** in the first direction D**1**. The bridge **324** may extend from a portion of the periphery **323***a* of the opening **323**. When the camera housing **320** is coupled to the bracket **330** in the first direction D**1**, the through hole **333** may be formed at a corner **330***b* in a direction opposite to the direction in which the camera housing **320** is inserted within the side surface of the bracket **330**, in order to correspond to the bridge **324** extending from a portion of the periphery **323***a* of the opening **323**. For example, the through hole **333** may be formed along a portion of the corner **330***b* in the fourth direction D**4**, which is opposite to the first direction D**1**, within the side surface of the bracket **330**. For example, the through hole **333** may extend from a portion of the corner **330***b* of the fourth direction D**4** within the side surface of the bracket **330** to the first direction D**1**. The bridge **324** may be exposed to the outside of the bracket **330**, through the through hole **333**. The bridge **324** passing the through hole **333** may be electrically connected to a printed circuit board (e.g., the printed circuit board **310** of FIG. **4**) and/or a support (e.g., the support **344** of FIG. **4**). When the camera housing **320** includes a plurality of bridges **324**, the bracket **330** may include a plurality of through holes **333** corresponding to each of the plurality of bridges **324**. The plurality of through holes **333** may be formed along a portion of the corner **330***b* of the fourth direction D**4** within the side surface of the bracket **330**, respectively.

[0114] Referring to FIG. **12**B, the camera housing **320** may be inserted into the bracket **330** in the fourth direction D**4**. The bridge **324** may extend from a portion of the periphery **323***a* of the opening **323**. When the camera housing **320** is coupled to the bracket **330** in the fourth direction D**4**, the through hole **333** may be formed in the corner **330***c* in a direction opposite to the direction in which the camera housing **320** is inserted within the side surface of the bracket **330**, in order to correspond to the bridge **324** extending from a portion of the periphery **323***a* of the opening **323**. For example, the through hole **333** may be formed along a portion of the corner **330***c* in the first direction D**1**, which is opposite to the fourth direction D**4**, within the side surface of the bracket **330**. For example, the through hole **333** may extend from a portion of the corner **330***c* in the first direction D**1** within the side surface of the bracket **330**, toward the fourth direction D**4**. The bridge **324** may be exposed to the outside of the bracket **330**, through the through hole **333**. The bridge **324** passing the through hole **333** may be electrically connected to the printed circuit board **310** and/or the support **344**. When the camera housing **320** includes a plurality of bridges **324**, the bracket **330** may include a plurality of through holes **333** corresponding to each of the plurality of bridges **324**. The plurality of through holes **333** may be formed along a portion of the corner **330***c* of the first direction D**1** within the side surface of the bracket **330**, respectively.

[0115] Referring to FIG. **12**C, the camera housing **320** may be inserted into the bracket **330** in the fourth direction D**4**. The bridge **324** may be extended from the second surface **322** of the camera housing **320**. For example, the bridge **324** may be extended from a portion of the other end portion **322***b* of the second surface **322** opposite to the end portion **322***a* of the second surface **322** connected to the first surface **321**. When the camera housing **320** is coupled to the bracket **330** in the fourth direction D**4**, the through hole **333** may be formed in the corner **330***c* in a direction opposite to the direction in which the camera housing **320** is inserted within the side surface of the bracket **330**, in order to correspond to the bridge **324** extending from a portion of the other end portion **322***b* of the second surface **322**. For example, the through hole **333** may be formed along a portion of the corner **330***c* in the first direction D**1**, which is opposite to the fourth direction D**4**, within the side surface of the bracket **330**. For example, the through hole **333** may extend from a portion of the corner **330***c* in the first direction D**1** within the side surface of the bracket **330**, toward the fourth direction D**4**. The bridge **324** may be exposed to the outside of the bracket **330**, through the through hole **333**. The bridge **324** passing the through hole **333** may be electrically connected to a printed circuit board (e.g., the printed circuit board **310** of FIG. **4**) and/or a support (e.g., the support **344** of FIG. **4**). When the camera housing **320** includes a plurality of bridges **324**, the bracket **330** may include a plurality of through holes **333** corresponding to each of the plurality of bridges **324**. The plurality of through holes **333** may be formed along a portion of the corner **330***c* of the first direction D**1** within the side surface of the bracket **330**, respectively. According to an embodiment, when the bridge **324** passes through the through hole **333** and is connected to the printed circuit board **310** and/or the support **344**, a position of the through hole **333** may be determined based on a coupling direction of the camera housing **320** relative to the bracket **330** and/or a position of the bridge **324**. FIGS. **12**A, **12**B, and **12**C are examples, and various embodiments may be possible.

[0116] FIG. **13**A is an exploded perspective view of an example camera **181** and camera housing **320** according to various embodiments. FIG. **13**B is a perspective view illustrating an injection portion **326** of an example camera housing **320** according to various embodiments. FIG. **14**A is a cross-sectional view of an example electronic device **101** taken along line A-A′ of FIG. **3** according to various embodiments. FIG. **14**B is a diagram illustrating an enlarged view of a portion X of FIG. **14**A according to various embodiments.

[0117] Referring to FIGS. **13**A and **13**B, the camera housing **320** may include an injection portion **326** and a connecting member **328**. For example, the injection portion **326** may be disposed inside the camera housing **320**. For example, the injection portion **326** may be contacted with an inner surface of the camera housing **320**. The injection portion **326** may be configured to support one or more components of the camera module **180** inside the camera housing **320**. For example, the injection portion **326** may at least partially surround the inner surface of the camera housing **320**. For example, the injection portion **326** may support a lens **210** positioned inside the camera housing **320** and/or an actuator (not illustrated) for driving the lens **210**, and may protect the lens **210** and/or the actuator. For example, the injection portion **326** may include a non-conductive material (e.g., polymer).

[0118] According to an embodiment, the camera housing **320** at least partially surrounding the camera **181** and the injection portion **326** may include an opening for the camera **181**. For example, referring to FIG. **13**A, a first opening **323** for the lens **210** exposed to the outside of the camera housing **320** may be formed in the first surface **321** of the camera housing **320**. For example, referring to FIG. **13**B, the injection portion **326** may include a second opening **326***a* for electrical connection between the camera **181** and a printed circuit board (e.g., the printed circuit board **310** of FIG. **14**A). The second opening **326***a* may be formed on a surface (e.g., bottom surface) of the injection portion **326** opposite the first surface **321**. For example, the first surface **321** may be a surface facing the first direction D**1**. For example, the injection portion **326** may form at least a portion of a surface facing the fourth direction D**4**.

[0119] According to an embodiment, the connecting member **328** may electrically connect the camera housing **320** to a ground portion of a printed circuit board **310** (e.g., printed circuit board **310** of FIG. **14**A). The camera housing **320** may be grounded via the connecting member **328**. Referring to FIG. **13**B, the connecting member **328** may be positioned within a groove **327** of the injection portion **326**. For example, the groove **327** may be formed by cutting a portion of the injection portion **326** from a periphery of the second opening **326***a* toward an inner surface of the camera housing **320**.

[0120] For example, the groove **327** may include a plurality of grooves **327***a*, **327***b*, and **327***c* that are spaced apart from each other. As illustrated in FIG. **13**B, the groove **327** may include a first groove **327***a*, a second groove **327***b*, and/or a third groove **327***c* that are spaced apart from each other, but the disclosure is not limited thereto. Since the injection portion **326** contacts the inner surface of the camera housing **320**, the inner surface of the camera housing **320** contacted with the injection portion **326** may not be exposed by the injection portion **326**. Since the groove **327** is formed by cutting a portion of the injection portion **326**, a portion of the inner surface of the camera housing **320** may be exposed through the groove **327**.

[0121] For example, the connecting member **328** may be positioned within the groove **327**. The connecting member **328** may include a plurality of connecting members **328** that contact a plurality of portions of the inner surface of the camera housing **320** exposed through the plurality of grooves **327***a*, **327***b*, and **327***c*. For example, the connecting member **328** may include a first connecting member **328***a* positioned within the first groove **327***a*, a second connecting member **328***b* positioned within the second groove **327***b*, and/or a third connecting member **328***c* positioned within the third groove **327***c*, but the disclosure is not limited thereto. For example, positions and numbers of the grooves **327** and the connecting members **328** may vary according to the size of the camera module **180** and the design of the electronic device **101**.

[0122] Referring to FIG. **14**A, the camera housing **320** may be electrically connected to the ground portion of the printed circuit board **310** through the connecting member **328**. For example, the camera **181** and the camera housing **320** may be disposed on the printed circuit board **310**. For example, the connecting member **328** may extend from the inner surface of the camera housing **320** exposed through the groove **327** to the ground portion of the printed circuit board **310**. Although not illustrated, the ground portion may include a ground layer of the printed circuit board **310**. Since the groove **327** is formed by cutting a portion of the injection portion **326**, a portion of the inner surface of the camera housing **320** corresponding to the groove **327** may not be covered by the injection portion **326**. The camera housing **320** may be electrically connected to the ground portion through the connecting member **328** that contacts the inner surface of the camera housing **320** exposed through the groove **327** and the ground portion of the printed circuit board **310**. The connecting member **328** may include a conductive material (e.g., metal) to electrically connect the camera housing **320** and the ground portion. For example, the connecting member **328** may be implemented as a portion of a c-clip, but is not limited thereto.

[0123] Referring to FIG. **14**B, the connecting member **328** that contacts the inner surface of the camera housing **320** and the ground portion of the printed circuit board **310**, may not be exposed to the outside of the camera housing **320**. For example, the connecting member **328** may be overlapped with the camera housing **320** by extending from a portion the inner surface of the camera housing **320** exposed through the groove **327** within the camera housing **320**. The connecting member **328** being overlapped with the camera housing **320** may indicate that the connecting member **328** is positioned entirely inside the camera housing **320**, without a portion positioned outside the camera housing **320**. For example, a shape of the connecting member **328** may be a shape having an approximately semicircular cross-section, unlike a shape of a typical c-clip having an approximately circular cross-section, but the disclosure is not limited thereto. For example, the connecting member **328** may have a partially bent shape.

[0124] According to an embodiment, since the connecting member **328** electrically connects the camera housing **320** and the ground portion within the camera housing **320**, a region of the printed circuit board **310** for the connecting member **328** exposed to the outside of the camera housing **320** may not be required. For example, when a portion of the connecting member **328** is positioned outside the camera housing **320**, a region of the printed circuit board **310** for soldering the portion of the connecting member **328** exposed to the outside of the camera housing **320** to the printed circuit board **310** may be required. Since the region of the printed circuit board **310** for the soldering is required, a size of the printed circuit board **310** may be increased. According to an embodiment, since the connecting member **328** extends inside the camera housing **320** and is positioned inside the housing **320**, the camera housing **320** may be electrically connected to the ground portion without increasing the size of the printed circuit board **310**. According to an embodiment, since the camera housing **320** is directly grounded to the ground region through the connecting member **328**, the grounding performance of the camera housing **320** may be enhanced.

[0125] According to an example embodiment, an electronic device (e.g., the electronic device **101** of FIG. **3**) may comprise a printed circuit board (e.g., the printed circuit board **310** of FIG. **6**), a camera (e.g., the camera **181** of FIG. **6**), a camera housing (e.g., the camera housing **320** of FIG. **6**), and a bracket (e.g., the bracket **330** of FIG. **6**). The printed circuit board may include a ground portion. The camera may include a lens (e.g., the lens **210** of FIG. **6**). The lens may face a first direction. The camera housing may include an opening (e.g., the opening **323** of FIG. **6**). The opening may be configured to expose the lens to the outside of a first surface (e.g., first surface **321** of FIG. **6**) of the camera housing facing the first direction. The camera housing may surround at least a portion of the camera. The bracket may include a third surface (e.g., the third surface **331** of FIG. **6**). The third surface may face away from a second surface (e.g., the second surface **322** of FIG. **6**) of the camera housing perpendicular to the first surface. The bracket may be electrically connected to the ground portion. The bracket may at least partially surround the camera housing. The camera housing may further include a bridge (e.g., the bridge **324** of FIG. **6**). The bridge may extend from the camera housing to the third surface of the bracket.

[0126] According to an example embodiment, the bridge may extend from a portion of a periphery (e.g., the periphery **323***a* of FIG. **5**A) of the opening to the third surface of the bracket. The bridge may include a bending portion (e.g., the bending portion **324***a* of FIG. **6**). The bending portion may be bent from the portion toward a third direction opposite to a second direction toward the lens passing through the opening. According to an embodiment of the present disclosure, the camera housing may be electrically connected to a ground portion through a bridge. The bridge may contact a bracket electrically connected to the ground portion. As the camera housing is electrically connected to the ground portion, the camera housing may be grounded. Since the camera housing surrounds at least a portion of the camera, it may be configured to shield electromagnetic waves reaching the camera from other electronic components and/or electromagnetic waves emitted from the camera. As the camera housing is grounded, electromagnetic interference between the camera and other electronic components (e.g., speaker, microphone, or antenna) within the electronic device may be reduced. According to an embodiment, the electronic device may reduce malfunctions and/or degradation of signal quality due to noise by reducing electromagnetic interference. The bridge for grounding the camera housing may fasten the camera housing to the bracket by supporting the camera housing with respect to the bracket. Since the bridge extends from a periphery of the opening within the first surface, the stress generated by supporting the camera housing may not be transferred to the camera inside the camera housing, but may transferred to the first surface. According to an embodiment, the stress applied to the camera within the camera housing may be reduced.

[0127] According to an example embodiment, the camera housing may be electrically connected to the ground portion, through the bridge contacted with the third surface of the bracket, which is electrically connected to the ground portion. According to an example embodiment of the present disclosure, the bridge may electrically connect the camera housing and the bracket by contacting the bracket. Since the bracket is electrically connected to the ground portion, the camera housing may be grounded through the bracket. According to an example embodiment, the camera housing may be grounded through a structure of the camera housing without a separate connecting member (e.g., c-clip or conductive tape) for grounding the camera housing.

[0128] According to an example embodiment, the bridge may extend to the third surface through a gap (e.g., the gap g of FIG. **6**) between the second surface of the camera housing and the third surface of the bracket. According to an example embodiment of the present disclosure, a gap may be formed between the camera housing and the camera. The bridge may support the camera housing relative to the bracket by extending through the gap. In the gap between the second surface

and the third surface, the bridge may fix the camera housing within the bracket.
[0129] According to an example embodiment, the bridge may include a first portion (e.g., the first portion **324-1** of FIG. **6**) and a second portion (e.g., the second portion **324-2** of FIG. **6**). The first portion may extend from the bending portion in the third direction. The second portion may extend from the first portion in a direction between the third direction and a fourth direction opposite to the first direction. The second portion may be contacted with the third surface.
[0130] According to an example embodiment, at least a portion of the second portion may extend to a gap between the second surface and the third surface.
[0131] According to an example embodiment, the camera may be coupled to the camera housing in the fourth direction. According to an example embodiment of the present disclosure, the bridge may include a first portion extending from a periphery of the opening toward a periphery of the first surface and a second portion extending from the first portion to the third surface of the bracket. The first portion and the second portion may electrically connect the camera housing to the bracket while occupying a minimum space. The camera housing including the bending portion, the first portion, and the second portion may be coupled to the bracket in a fourth direction opposite the first direction.
[0132] According to an example embodiment, the bridge may include a first portion (e.g., the first portion **324-1** of FIG. **8**A), a second portion (e.g., the second portion **324-2** of FIG. **8**A), and a third portion (e.g., the third portion **324-3** of FIG. **8**A). The first portion may extend from the bending portion in the third direction. The second portion may extend, from the first portion, in a direction between the third direction and a fourth direction opposite to the first direction, to a gap between the second surface and the third surface. The third portion may extend from the second portion in a direction between the first direction and the third direction. The third portion may be contacted with the third surface.
[0133] According to an example embodiment, the camera may be coupled to the camera housing in the first direction. According to an example embodiment of the present disclosure, the bridge may include a bending portion, a first portion, a second portion, and a third portion. The camera housing including the bridge of the above structure may be coupled to the bracket in the first direction. According to an example embodiment, the structure of the bridge may be determined based on a coupling direction of the camera housing and the bracket. According to an example embodiment, the camera housing may be electrically connected to the bracket by including a bridge of an appropriate structure according to the coupling direction of the camera housing and the bracket.
[0134] According to an example embodiment, the bracket may include a groove (e.g., the groove **332** of FIG. **6**). The groove may be formed within the third surface. The bridge may include an inserting portion (e.g., the inserting portion **324***b* of FIG. **6**). The inserting portion may be inserted into the groove. The inserting portion may fix the camera housing to the bracket. The camera housing may be fixed within the bracket through the inserting portion inserted into the groove. The inserting portion may prevent and/or suppress the camera housing from being separated from the bracket. The contact area between the bridge and the bracket may be increased through the inserting portion and the groove that are contacted with each other. As the contact area between the bridge and the bracket is increased, an electrical connection between the camera housing and the bracket may be improved, so the grounding effect of the camera housing may be improved.
[0135] According to an example embodiment, the bridge may elastically support the camera housing with respect to the bracket. According to an embodiment of the present disclosure, the bridge may have elasticity. For example, the bridge may include an elastic material. The bridge may be inserted into the gap between the second surface and the third surface in a compressed state. By the elasticity of the bridge, the camera housing may be coupled within the bracket. The bridge may connect the camera housing and the bracket without a separate coupling member, by coupling the camera housing to the bracket. Since the bridge has elasticity, the stress applied to the camera housing may be reduced. The bridge having elasticity may reduce damage to the camera

within the camera housing by reducing the stress.
[0136] According to an example embodiment, the camera housing may include a plurality of bridges (e.g., the plurality of bridges **325***a*, **325***b*, **325***c*, and **325***d* of FIG. **5**B). The plurality of bridges may be formed along the periphery of the opening. The plurality of bridges may be symmetrical to each other. According to an example embodiment of the present disclosure, the plurality of bridges may be disposed symmetrically to each other. The plurality of bridges may increase the grounding area for grounding the camera housing. Through the plurality of bridges, the grounding effect of the camera housing may be improved.
[0137] According to an example embodiment, the bridge may penetrate the bracket. The bridge may extend to the ground portion. According to an embodiment of the present disclosure, the bridge may be directly connected to the ground portion. As the bridge is directly connected to the ground portion, the grounding effect of the camera housing may be improved.
[0138] According to an example embodiment, the electronic device may further include a support (e.g., the support **344** of FIG. **11**B). The support may support a portion of components of the electronic device. The support may include a grounding region. The bridge may penetrate the bracket. The bridge may extend to the grounding region of the support. According to an embodiment of the present disclosure, the bridge may be directly connected to the grounding region of the support. As the bridge is directly connected to the grounding region, the grounding effect of the camera housing may be improved.
[0139] According to an example embodiment, the bracket may include a through hole (e.g., the through hole **333** of FIG. **11**A) and a protruding portion (e.g., the protruding portion **334** of FIG. **11**A). The bridge may pass the through hole. The protruding portion may be contacted with the through hole. A screw (e.g., the screw **350** of FIG. **11**A) may be inserted into the protruding portion. The screw may penetrate the protruding portion and the bridge. The screw may be inserted into the grounding region to electrically connect the bridge to the grounding region. According to an example embodiment of the present disclosure, when the bridge penetrates the bracket, the bracket may include a through hole for the bridge. The bridge may be exposed to the outside of the bracket, through the through hole. The bridge exposed to the outside of the bracket may be connected to the grounding region of the support through the screw. The bracket may be fixed to the support through the screw penetrating the protruding portion.
[0140] According to an example embodiment, the first surface may include a cut portion (e.g., the cut portion **321***b* of FIG. **5**A) cut out from a portion of the periphery of the opening toward the periphery of the first surface, such that the bending portion is disposed outside the opening. According to an example embodiment of the present disclosure, since the bending portion extends from a portion of the periphery of the opening, it may protrude into the opening. When the bending portion protrudes into the opening, interference may occur with a lens of the camera. In order to eliminate the interference, the first surface may include a portion cut along a portion where the bridge is formed, toward the periphery of the first surface. When the bending portion is bent, the cut portion may be bent toward the periphery of the first surface. According to an embodiment, interference between the bending portion and the lens may be reduced because the bending portion does not protrude into the opening.
[0141] According to an example embodiment, a camera module (e.g., the camera module **180** of FIG. **3**) may include a camera (e.g., the camera **181** of FIG. **6**), a camera housing (e.g., the camera housing **320** of FIG. **6**), and a bracket (e.g., the bracket **330** of FIG. **6**). The camera may include a lens (e.g., the lens **210** of FIG. **6**). The lens may face a first direction. The camera housing may include an opening (e.g., the opening **323** of FIG. **6**). The opening may be formed for the lens exposed toward the outside within a first surface (e.g., the first surface **321** of FIG. **6**) facing the first direction. The camera housing may surround at least a portion of the camera. The bracket may include a third surface (e.g., the third surface **331** of FIG. **6**). The third surface may face away from a second surface (e.g., the second surface **322** of FIG. **6**) of the camera housing perpendicular to the

first surface. The bracket may be electrically connected to the ground portion of the printed circuit board. The bracket may at least partially surround the camera housing. The camera housing may further include a bridge (e.g., the bridge **324** of FIG. **6**). The bridge may extend from a portion of the periphery of the opening to the third surface of the bracket. The bridge may include a bending portion (e.g., the bending portion **324***a* of FIG. **6**). The bending portion may be bent from the portion in a third direction opposite to a second direction toward the lens passing through the opening. According to an example embodiment of the present disclosure, the camera housing may be electrically connected to the ground portion through a bridge. The bridge may be contacted with a bracket electrically connected to the ground portion. As the camera housing is electrically connected to the ground portion, the camera housing may be grounded. Since the camera housing surrounds at least a portion of the camera, it may be configured to shield electromagnetic waves reaching the camera from other electronic components and/or electromagnetic waves emitted from the camera. The bridge for grounding the camera housing may fasten the camera housing to the bracket by supporting the camera housing with respect to the bracket. Since the bridge extends from the periphery of the opening in the first surface, the stress generated by supporting the camera housing may not be transferred to the camera inside the camera housing but may be transferred to the first surface. According to an example embodiment, the stress applied to the camera inside the camera housing may be reduced.

[0142] According to an example embodiment, the camera housing may be electrically connected to the ground portion, through the bridge contacted with the third surface of the bracket electrically connected to the ground portion. According to an example embodiment of the present disclosure, the bridge may electrically connect the camera housing and the bracket by contacting the bracket. Since the bracket is electrically connected to the ground portion, the camera housing may be grounded through the bracket. According to an example embodiment, the camera housing may be grounded through the structure of the camera housing without a separate connecting member (e.g., c-clip or conductive tape) for grounding the camera housing.

[0143] According to an example embodiment, the bridge may extend to the third surface through a gap (e.g., the gap g of FIG. **6**) between the second surface of the camera housing and the third surface of the bracket. According to an example embodiment of the present disclosure, a gap may be formed between the camera housing and the camera. The bridge may support the camera housing relative to the bracket by extending through the gap. In the gap between the second surface and the third surface, the bridge may fix the camera housing within the bracket.

[0144] According to an example embodiment, the bridge may include a first portion (e.g., the first portion **324-1** of FIG. **6**) and a second portion (e.g., the second portion **324-2** of FIG. **6**). The first portion may extend from the bending portion in the third direction. The second portion may extend from the first portion in a direction between the third direction and a fourth direction opposite to the first direction. The second portion may be contacted with the third surface. According to an example embodiment of the present disclosure, the bridge may include a first portion extending from the periphery of the opening toward the periphery of the first surface, and a second portion extending from the first portion to the third surface of the bracket. The first portion and the second portion may electrically connect the camera housing to the bracket while occupying a minimum space. The camera housing including a bending portion, a first portion, and a second portion may be coupled to the bracket in a fourth direction opposite to the first direction.

[0145] According to an example embodiment, the bridge may include a first portion (e.g., the first portion **324-1** of FIG. **8**A), a second portion (e.g., the second portion **324-2** of FIG. **8**A), and a third portion (e.g., the third portion **324-3** of FIG. **8**A). The first portion may extend from the bending portion in the third direction. The second portion may extend, from the first portion, in a direction between the third direction and a fourth direction opposite to the first direction, to a gap between the second surface and the third surface. The third portion may extend from the second portion in a direction between the first direction and the third direction. The third portion may be contacted with the third surface. According to an example embodiment of the present disclosure, the bridge may include a bending portion, a first portion, a second portion, and a third portion. The camera housing including the bridge of the above structure may be coupled to the bracket in the first direction. According to an example embodiment, the structure of the bridge may be determined based on the coupling direction of the camera housing and the bracket. According to an example embodiment, the camera housing may be electrically connected to the bracket by including a bridge of an appropriate structure according to the coupling direction of the camera housing and the bracket.

[0146] According to an example embodiment, the bracket may include a groove (e.g., the groove **332** of FIG. **6**). The groove may be formed within the third surface. The bridge may include an inserting portion (e.g., the inserting portion **324***b* of FIG. **6**). The inserting portion may be inserted into the groove. The inserting portion may fix the camera housing to the bracket. The camera housing may be fixed within the bracket through the inserting portion inserted into the groove. The inserting portion may prevent and/or suppress the camera housing from being separated from the bracket. The contact area between the bridge and the bracket may be increased through the inserting portion and the groove that are contacted with each other. As the contact area between the bridge and the bracket is increased, the electrical connection between the camera housing and the bracket may be improved, so the grounding effect of the camera housing may be improved.

[0147] According to an example embodiment, an electronic device (e.g., the electronic device **101** of FIG. **3**) may include a printed circuit board (e.g., the printed circuit board **310** of FIG. **10**A), a camera (e.g., the camera **181** of FIG. **10**A), a camera housing (e.g., the camera housing **320** of

[0148] FIG. **10**A), and a bracket (e.g., the bracket **330** of FIG. **10**A). The printed circuit board may include a ground portion. The camera may include a lens (e.g., the lens **210** of FIG. **10**A) facing a first direction. The camera housing may include an opening (e.g., the opening **323** of FIG. **10**A) for the lens exposed to the outside within a first surface (e.g., the first surface **321** of FIG. **10**A) facing the first direction. The camera housing may surround at least a portion of the camera. The bracket may include a third surface (e.g., the third surface **331** of FIG. **10**A). The third surface may face away from a second surface (e.g., the second surface **322** of FIG. **10**A) of the camera housing perpendicular to the first surface. The bracket may be electrically connected to the ground portion. The bracket may at least partially surround the camera housing. The camera housing may further include a bridge (e.g., the bridge **324** of FIG. **10**A). The bridge may extend, from a portion of another end portion (e.g., the other end portion **322***b* of FIG. **10**A) of the second surface opposite to an end portion (e.g., the end portion **322***a* of FIG. **10**A) of the second surface connected to the first surface, to the third surface of the bracket. The bridge may include a first portion (e.g., the first portion **324-1** of FIG. **10**A). The first portion may extend in a direction between the first direction and a third direction opposite to a second direction toward the lens passing through the opening from the first portion.

[0149] According to an example embodiment, the bridge may further include a second portion (e.g., the second portion **324-2** of FIG. **10**B). The second portion may extend, from the first portion, in a direction between the third direction and a fourth direction opposite to the first direction. The second portion may be contacted with the third surface.

[0150] According to an example embodiment, an electronic device may comprise a printed circuit board, a camera, a camera housing, an injection portion (e.g., the injection portion **326** of FIG. **13**B), and a connector (e.g., the connecting member **328** of FIG. **13**B). The printed circuit board may include a ground portion. The camera may include a lens. The camera may be disposed on the printed circuit board. The camera housing may surround at least a portion of the camera. The injection portion may be disposed inside the camera housing. The injection portion may surround at least a portion of the camera. The connector may extend from a portion of the inner surface of the camera housing exposed through a groove (e.g., the groove **327** of FIG. **13**B) of the injection portion, to the ground portion. According to an embodiment of the present disclosure, the camera

housing and the ground portion may be electrically connected through the connector. The connector may contact the inner surface of the camera housing exposed through the groove. Since the camera housing may be directly grounded, the ground performance of the camera module may be enhanced. As the ground performance of the camera module is enhanced, electromagnetic interference (EMI) and/or electromagnetic susceptibility (EMS) may be improved.
[0151] According to an example embodiment, the connector may overlap the camera housing, by extending from the portion of the inner surface to the ground portion within the camera housing. According to an example embodiment of the present disclosure, the connector may be positioned only inside the camera housing. Since the connector is not exposed to the outside of the camera housing, an increase in the size of the printed circuit board may not be required.
[0152] According to an example embodiment, the groove may include a plurality of grooves spaced apart from each other. The connector may include a plurality of portions of the inner surface of the camera housing exposed through the plurality of grooves and a plurality of connectors contacting the ground portion. According to an example embodiment of the present disclosure, numbers and positions of the groove and the connector(s) may vary.
[0153] According to an example embodiment, the injection portion may include an opening for the camera.
[0154] According to an example embodiment, the groove may be formed by cutting a portion of the injection portion toward the inner surface of the camera housing, from the periphery of the opening. According to an example embodiment of the present disclosure, as a portion of the injection portion surrounding the inner surface of the camera housing is cut, the inner surface of the camera housing may be exposed. The connector may electrically connect the camera housing to the ground portion by contacting the exposed inner surface of the camera housing.
[0155] The problems addressed in the disclosure are not limited to those described above, and various modifications may be made without departing from the spirit and scope of the disclosure.
[0156] The effects that can be obtained from the present disclosure are not limited to those described above, and any other effects not mentioned herein will be clearly understood by those having ordinary knowledge in the art to which the present disclosure belongs, from the description.
[0157] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, a home appliance, or the like. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.
[0158] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “**1**st” and “**2**nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” or “connected with” another element (e.g., a second element), the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.
[0159] As used in connection with various embodiments of the disclosure, the term “module” may

include a unit implemented in hardware, software, or firmware, or any combination thereof, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0160] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between a case in which data is semi-permanently stored in the storage medium and a case in which the data is temporarily stored in the storage medium.

[0161] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0162] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0163] No claim element is to be construed under the provisions of 35 U.S.C. § 112, sixth paragraph, unless the element is expressly recited using the phrase “means for” or “means.”

[0164] While the disclosure has been illustrated and described with reference to various example embodiments, it will be understood that the various example embodiments are intended to be illustrative, not limiting. It will be further understood by those skilled in the art that various modifications, alternatives and/or variations of the various example embodiments may be made without departing from the true technical spirit and full technical scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

Claims

1. An electronic device comprising: a printed circuit board (PCB) including a ground portion; a camera including a lens facing a first direction; a camera housing, surrounding at least portion of the camera, including an opening configured to expose the lens to an outside of a first surface of the camera housing facing the first direction; and a bracket, including a third surface facing away from a second surface of the camera housing perpendicular to the first surface, electrically connected to the ground portion, and at least partially surrounding the camera housing, wherein the camera housing includes a bridge extending from the camera housing to the third surface of the bracket.
2. The electronic device of claim 1, wherein the bridge extends from a portion of a periphery of the opening to the third surface of the bracket, and wherein the bridge includes a bending portion bending from the portion of the periphery of the opening in a third direction, the third direction opposite to a second direction toward the lens passing through the opening.
3. The electronic device of claim 2, wherein the bridge extends to the third surface via a gap between the second surface of the camera housing and the third surface of the bracket.
4. The electronic device of claim 2, wherein the bridge includes: a first portion extending from the bending portion in the third direction, and a second portion, extending from the first portion in a direction between the third direction and a fourth direction opposite to the first direction, contacted with the third surface.
5. The electronic device of claim 4, wherein at least portion of the second portion extends to the gap between the second surface and the third surface.
6. The electronic device of claim 2, wherein the bridge includes: a first portion extending from the bending portion in the third direction, and a second portion, extending from the first portion, in a direction between the third direction and a fourth direction opposite to the first direction, to a gap between the second surface of the camera housing and the third surface of the bracket, and a third portion, extending from the second portion in a direction between the first direction and the third direction, contacted with the third surface.
7. The electronic device of claim 1, wherein the bracket includes a groove formed in the third surface, and wherein the bridge includes an inserting portion, configured to be inserted into the groove, fixing the camera housing to the bracket.
8. The electronic device of claim 1, wherein the bridge is configured to elastically support the camera housing with respect to the bracket.
9. The electronic device of claim 1, wherein the camera housing includes a plurality of bridges formed along the periphery of the opening, and wherein the plurality of bridges are symmetrical to each other.
10. The electronic device of claim 1, wherein the bridge penetrates the bracket and extends to the ground portion.
11. The electronic device of claim 1, further comprising a support, configured to support a portion of components of the electronic device, including a ground region, wherein the bridge penetrates the bracket and extends to the ground region of the support.
12. The electronic device of claim 1, wherein the bracket includes: a through hole through which the bridge is configured to pass, and a protruding portion, contacting the through hole, configured to receive a screw, and wherein the screw is configured to electrically connect the bridge to the ground portion by penetrating the protruding portion and the bridge and being inserted into the ground portion.
13. The electronic device of claim 2, wherein the first surface includes a cut portion cut out from a portion of the periphery of the opening toward the periphery of the first surface, such that the bending portion is disposed outside the opening.

14. The electronic device of claim 1, wherein the bridge extends from a portion of an end portion of the second surface opposite to an end portion of the second surface connected to the first surface, to the third surface of the bracket, and wherein the bridge includes a first portion extending from the portion in a direction between the first direction and a third direction, the third direction opposite to a second direction toward the lens passing through the opening.

15. The electronic device of claim 14, wherein the bridge includes a second portion, contacting the third surface, extending from the first portion in a direction between the third direction and a fourth direction, the fourth direction opposite to the first direction.

16. The electronic device of claim 1, wherein the camera housing is electrically connected to the ground portion, via the bridge contacting the third surface of the bracket electrically connected to the ground portion.

17. A camera module comprising: a camera including a lens facing a first direction; a camera housing surrounding at least portion of the camera, the camera housing including: a first surface facing the first direction and defining an opening, wherein the lens is exposed to an outside of the camera housing via the opening, and a second surface perpendicular to the first surface of the camera housing; and a bracket, at least partially surrounding the camera housing, including a third surface facing the second surface of the camera housing, and electrically connected to the ground portion of the PCB, wherein the camera housing includes a bridge extending from the camera housing to the third surface of the bracket, and wherein the camera housing is electrically connected to the ground portion, via the bridge contacting the third surface of the bracket electrically connected to the ground portion of the PCB.

18. The camera module of claim 17, wherein the bridge extends from a portion of a periphery of the opening to the third surface of the bracket, and wherein the bridge includes a bending portion bending from the portion of the periphery of the opening in a third direction, the third direction being opposite to a second direction toward the lens passing through the opening.

19. The camera module of claim 18, wherein the bridge includes: a first portion extending from the bending portion in the third direction, and a fourth portion, extending from the first portion in a direction between the third direction and a fourth direction opposite to the first direction, contacted with the third surface of the bracket.

20. The camera module of claim 17 wherein the bridge is configured to elastically support the camera housing with respect to the bracket.
